

cs-mail: Express Lanes for Bounded Communication

A Pedagogical Memo

C-SQD

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Nonnormative memo. This document explains a concept and motivates it for a reader new to the underlying ideas. It introduces *express lanes*, the user-facing form of the scoped capabilities described in the *cs-mail Protocol Specification*. The settlement and privacy semantics remain governed by that specification. The companion implementation plan develops issuance and lifetime; the deployment profile defines the separate C-SQD financial program.

Current account scope. The current C-SQD product has one identifiable person per account, tied to a verified bank account, with multiple email addresses permitted. Organizational senders in these examples are legacy senders or possible future native protocol participants; they do not imply organizational C-SQD accounts. The [C-SQD domain model](#) controls that product distinction.

1 A First Week with a New Bank Account

You opened a checking account on Monday. By Friday your inbox holds a welcome note from `welcome@bank.com`, a statement notice from `statements@bank.com`, a fraud alert from `alerts@e.bank.com`, a marketing message you did not really ask for, and—indistinguishable at a glance—a message from “Bank Security Team” `<security@bank-support-online.com>` urging you to confirm your login. The first three you wanted. The last is a phishing attempt. Every one of them arrived the same way: someone typed your address and pressed send.

In *cs-mail*, a stranger selects a published request class and authorizes one conditional charge. You can accept with standing permission or a lane, or reject. Permitted follow-ups share the charge and deadline. Your bank needs a different invitation: you chose to hear from it for particular reasons, across many addresses. A bounded lane expresses that permission.

2 The Choice: Standing or Bounded Permission

Standing and bounded permission both apply to people and organizations: a friend may receive standing permission, a researcher an express lane. Familiar bounded uses include the bank’s statements, the airline’s boarding pass, the pharmacy’s refill notice, the clinic’s appointment reminder. Call this *transactional mail*.

Standing acceptance alone fits transactional mail poorly, for four reasons:

- It arrives from many addresses. One bank sends from `statements@`, `alerts@`, and `no-reply@e.bank.com`. Accepting an address would mean accepting dozens, and missing the next one.
- It arrives at volume and without your reply. You will never hand-curate it.

- It is narrow. You want the bank’s fraud alerts; you did not necessarily invite its marketing.
- The sender usually is not on *cs-mail* yet. Your bank still sends ordinary email, and will for a long time.

So we need a permission path that grants a chosen correspondent a bounded, revocable, effortless right to reach you—one that resists impersonation and works *before* the sender adopts *cs-mail*. We call it an *express lane*. Before describing it, two pieces of background make it comprehensible.

3 What You Need to Know First

3.1 The “From” line is not identity

The sender shown on an email is like the return address handwritten on an envelope: a claim, not a proof. Anyone can write “Your Bank” as the sender name and `security@bank.com` as the address; nothing in ordinary email stops them. Phishing is precisely this move. An express lane therefore can never be granted to a display name or a typed address, because neither is evidence of who actually sent the message.

3.2 Mail can prove authorized domain use: SPF, DKIM, and DMARC

Two widely deployed standards let a receiving system check a message cryptographically, and most newcomers have never heard of them.

DKIM attaches a domain signature to a message; *SPF* authenticates the domain responsible for the SMTP sending path. Either can authenticate a domain, but that domain is not necessarily the one visible to the reader.

DMARC, standardized by RFC 9989, ties a passing SPF or DKIM identity to the visible RFC 5322 author domain through strict or relaxed alignment. It also lets a domain publish its requested handling for failures.

Together they answer one precise question: did the owner of the visible author domain authorize its use on this message? They do not say the message is friendly, wanted, legally from a particular company, or safe. A look-alike domain can pass DMARC for itself. The lane’s separate expected-domain match supplies that missing decision.

3.3 What *cs-mail* already provides

cs-mail already gives us relationship requests for strangers and the idea of recipient-granted permission. The protocol specification also already defines a general *scoped capability*—a recipient-signed grant that lets a sender communicate without a request charge for a bounded purpose. Express lanes are that launch capability, given the issuance, safety, and lifetime rules it needs to actually work.

4 The Solution: Express Lanes

An express lane is a recipient-granted, revocable path without a relationship-request charge for one sender you have chosen, bounded in purpose and time. It concerns the relationship and is usually not scoped to a particular conversation. Three properties make it useful.

What it limits. A lane controls *who* may reach you, *how much*, and for *how long*. It can also constrain the authenticated purpose and automation mode that a sender is allowed to declare. Because *cs-mail* content is end-to-end encrypted, no provider reads a message to confirm it is “really about your account” or that a human wrote it. The lane enforces the sender’s accountable declaration, not the semantic truth of the ciphertext. If a sender uses that scope dishonestly, you can report it and revoke or block the lane. This is a deliberate limit, and an honest one.

What it binds to. For a sender already on *cs-mail*, the lane binds to the sender’s cryptographic protocol identity, and the provider can honor it without even learning which organization it is. For a legacy sender—your bank today—the lane binds to the canonical *organizational domain* established by a DMARC pass, `bank.com`, and honors a message only when its visible author domain falls under that identity. It never binds to a display name or a single address. So the same lane behaves sensibly across every case in the opening story:

- the fraud alert from `alerts@e.bank.com` and the statement from `statements@bank.com` both ride the lane, because both authentically come from `bank.com`;
- “Bank Security Team” `<security@bank-support-online.com>` does not, because its authenticated domain is not `bank.com`;
- `security@bank.com.evil.com` does not, because its authenticated domain is `evil.com`.

The phishing message from Friday fails at exactly this line.

Why it is safe to grant freely. An express lane is trust you can withdraw in one tap. If it is abused—the bank floods you, or a lane is misused—you can revoke it. Future messages then need another valid permission path. A sender with an open relationship request may send only the follow-ups your settings allow; otherwise a new request requires eligibility and explicit sender authorization. Revocation never silently charges the sender or settles an existing request. A block always overrides a lane.

These permission paths explain the main cases:

Who	How they reach you	How it is granted and ended
A stranger	Selects a request class and submits one charged request, with permitted follow-ups.	Explicit sender authorization; decision or expiry settles the request.
A standing contact	Writes with no request charge.	You grant standing permission; you revoke it.
A bounded contact	Travels an express lane with no request charge, bounded in purpose and time.	You grant a lane, including through request acceptance; revoke it or let it lapse.

The legacy binding is a bridge. Binding to a domain is a bridge, not the destination. The day your bank adopts *cs-mail*, its lane re-binds to the bank’s cryptographic identity, the DMARC dependency falls away, and the lane becomes fully private—the provider no longer needs to see `bank.com` to honor it. The bridge advertises its own obsolescence.

5 How You Get One: Issuance

An express lane should feel like a consequence of a decision you already made, never a settings screen you must remember to visit. Three useful moments are:

1. **When you hand over your address.** Signing up, checking out, or booking—the moment you give the bank your address, the client offers, in one line, to open an express lane for it.
2. **When you write first.** Emailing the bank yourself auto-opens a return lane, since you have plainly invited a reply.
3. **When you accept a request with bounded permission.** Professor Bob may welcome Alice’s research proposal with a purpose- or time-limited relationship lane. Acceptance commits the lane and full request refund obligation together, without granting standing permission.

The class determines collateral; the lane states granted permission. Both are implemented in the Rust libraries. A separate lane-management grant also resolves a pending request with the full refund; already-terminal settlements stay fixed.

For a legacy sender the grant cannot complete on a promise; it completes on proof. When you sign up on `bank.com`, the client remembers that you were on `bank.com` and expects mail from that domain. The first message that arrives claiming to be the bank is checked with DMARC; if it authentically comes from the expected domain, the client shows a one-tap confirmation naming the *authenticated domain*—not the display name—and the lane opens. Seeding the expectation from the website you were actually on turns “which bank?” into a simple match, and lets you catch a look-alike domain at the very moment of granting.

6 How a Lane Ends: Lifetime and Renewal

A lane’s lifetime describes the permission you gave an organization. Your annual utility charge describes the service you purchased from C-SQD. A lane carries no request charge, but delivery still requires funded service coverage. Renewing service does not extend the lane’s signed validity; a lapse in service coverage does not itself revoke your grant.

Within that distinction, the lane needs its own lifetime rules. A grant that lasted forever by default would leave forgotten permissions in place, so its ordinary horizon follows activity and its hard limit remains recipient-signed.

Anchor on activity, not the calendar. A lane’s clock is its last authenticated message, not its issue date. A lane in active use stays open on its own; one that goes quiet drifts toward expiry. This captures most of the benefit of tying a lane to an event—a single flight, say—without asking anyone to define events.

The default is a gentle re-confirm. When an inactive lane reaches the end of its window, you get a light prompt—keep the airline’s lane, or let it lapse?—rather than silent deletion. You set the window per sender.

The exception that matters. Some lanes should effectively never lapse. Your bank’s fraud alerts and your hospital’s test results are exactly the mail whose silent disappearance is most costly: a lapsed lane there means a missed fraud warning or a missed result, unable to use the lane

until authority is renewed. A new charge is never automatic. For these you choose a *persistent* lane: activity renews its ordinary horizon, and an inactive horizon produces a review reminder instead of a lapse. Even a persistent lane has a recipient-signed hard validity bound, after which a fresh grant is required.

Revocation and blocking remain available and immediate, whatever the lifetime.

7 Implementation Constraints

The following constraints shape any implementation. Their details belong in a later specification; the constraints themselves are worth stating now because they are what make the concept real.

Lanes live at the recipient’s provider. Only the provider sees inbound mail and runs the DMARC check, and the provider is already where an incoming message is evaluated for accepted, lane, or open-request permission before any new-request flow. The lane must be evaluated there. A device cannot verify a domain on mail it never handled.

The list of lanes is sensitive and must be compartmentalized. Which organizations a person trusts, and for what—their bank, their hospital, their lawyer—is precisely the relationship profile that *cs-mail*’s privacy model exists to protect. Lanes must therefore live in the relationship privacy domain, under their own keys and access role, not in a general table that a ledger or telemetry breach could read. For senders on *cs-mail*, a lane can be a recipient-signed, pairwise-scoped grant that the provider verifies without learning the organization at all; for legacy senders the provider must see the domain to match DMARC, which is one more reason the native path is the more private destination.

Lanes use the existing permission rules. A scoped capability already authorizes communication without a request charge, solicitation, or new-request history increment. Blocking overrides it. Express lanes give that capability a practical issuance flow, domain binding for legacy senders, and activity-based lifetime with an optional persistent mode. A lane grant creates no charge. Accepting a request with a lane also settles that request; later lane expiry or revocation does not reopen the settled money.

Declarations remain minimal. The authenticated envelope exposes only the coarse purpose, origin mode, validity, and capability reference required for admission or recipient-controlled routing. Structured message fields remain encrypted. A native sender signs its own declarations; a legacy gateway identifies itself as the declaration authority and does not turn DMARC into a claim about purpose or human authorship. These metadata receive the same minimization, compartmentalization, and retention discipline as other relationship signals.

Lane activity and member activity are different. An authenticated incoming message can renew a lane’s horizon. It does not by itself establish intentional activity for an annual member allocation, renew paid service, extend a pending relationship request, or create a pool contribution. Eligibility for an allocation depends on the completed year’s membership rules, not renewal of either service or a lane. Those lifecycles remain separate even when their events occur close together.